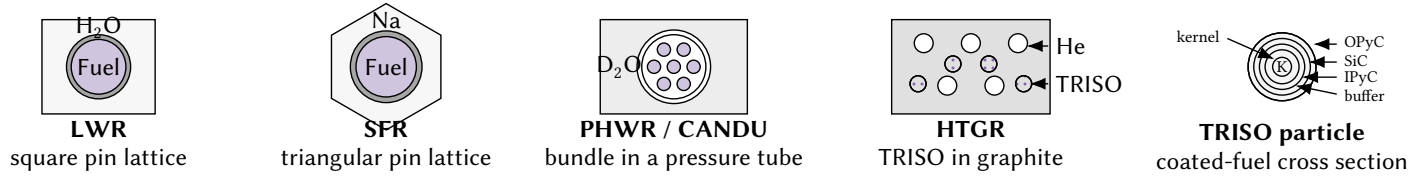


From mixtures to reactor lattices

Why should moving the same atoms into separate regions change multiplication?

15
LESSON

Read: §§ 4.1–4.2; Figs. 4.1–4.3 and Table 4.1.



Unit cells adapted from Fig. 4.1, with an added TRISO zoom. Fuel is shaded purple; the stippled HTGR compacts suggest dispersed TRISO particles. These sketches are not on the same scale.

1 Core design balances criticality and heat removal

Criticality over the fuel lifetime and removal of fission heat jointly determine the best lattice of unit cells. Identify **fuel, coolant, moderator, cladding/structure, and absorbers**; one material may serve more than one role.

$$P_{th} = \bar{P}'''V, \quad q' = \frac{P_{pin}}{L}, \quad k = k_{\infty}P_{NL}.$$

Thin fuel elements shorten heat-conduction distances; assemblies make handling and refuelling practical. A high k_{∞} alone does not establish an acceptable power density or a critical finite core.

Family	Distinction to find in the reading
PWR/BWR	H ₂ O cools and moderates; enriched UO ₂ . PWR: liquid primary loop and steam generator. BWR: boiling in the core.
PHWR	D ₂ O moderator surrounds pressure tubes; low absorption permits natural-U fuel.
Graphite	Separate moderator and coolant: He in an HTGR; TRISO particles are embedded in graphite-based fuel compacts or pebbles; boiling H ₂ O in an RBMK.
Fast	No intentional moderator; Na (SFR) or He (GCFR) coolant; tightly packed, enriched fuel.

2 Reduce the geometry deliberately

Core → assembly → repeating cell. Use a square or hexagonal cell to represent a uniform, repeated lattice. Slabs represent layers, cylinders represent pins, and spheres represent local lumps

3 The uranium-lump experiment

Begin with H₂O and natural U, then segregate U into a central metal sphere of radius r_U inside a water sphere of outer radius R . Let $h = H/U$ be the **total atom ratio**, not a local number-density ratio. With $V_w = V - V_U$,

$$h = \frac{N_H^w V_w}{N_U^{\text{metal}} V_U}, \quad r_U = R \left(\frac{N_H^w}{N_H^w + h N_U^{\text{metal}}} \right)^{1/3}.$$

Use nominal $\rho_U = 19.1 \text{ g/cm}^3$ for this model and the chosen water density:

$$N_U^{\text{metal}} = \frac{\rho_U N_A}{M_U}, \quad N_H^w = \frac{2\rho_w N_A}{M_{\text{H}_2\text{O}}}.$$

Here N_A is Avogadro's constant and M denotes molar mass. For the homogenized model, distribute the *same isotope inventories* over V :

$$\bar{N}_i = \frac{N_i^{\text{metal}} V_U + N_i^w V_w}{V}.$$

Spatial Self Shielding

Most slowing down can occur *outside* the fuel. Near U-238 capture resonances, absorption depresses the flux inside a lump, shielding its interior. This can improve resonance escape without adding fissile atoms. Thermal-flux depression competes with that benefit; larger lumps are not unconditionally better!

Lumping does *not* by itself make natural uranium in light water critical. Revisit that limitation when p and f are separated in Lessons 17–18.

OpenMC | Exploring the impact of lumping

Compare homogeneous and lumped models over H/U . Inspect η or k_{∞} , spectra, and U-238 capture.

Sources: supplied Ch. 4, §§ 4.1–4.2 and the self-shielding discussion in § 4.5; slide set 15. OpenMC reference: Geometry, § 6.1.1 (boundary conditions).